

Q-Describe a real-world scenario from any app you use (such as Zomato, Paytm, or IRCTC) where TACACS+ would be more suitable than RADIUS for managing admin access, and explain why.

Imagine **Zomato** has a data center containing many Cisco routers and switches. Network administrators need to log in to these devices to maintain the network.

In this situation, **TACACS+** would be more suitable than **RADIUS** for administrator access.

Scenario	TACACS+	Why it matters
Admin logs into a Cisco router	✓	TACACS+ authenticates the administrator
Different admins have different permissions	✓	Authorization can be customized per user
Allow only specific commands	✓	For example, one admin can use show commands but cannot change configuration
Track administrator actions	✓	Accounting can record commands and sessions
Secure management of network devices	✓	TACACS+ is designed particularly for device administration

Simple example

Suppose Zomato has:

- **Admin A:** Can view router configuration only
- **Admin B:** Can configure interfaces and routing
- **Admin C:** Full administrator access

With **TACACS+**, the organization can authenticate each admin and then authorize **specific commands** based on their role.

For example:

Admin A → login → TACACS+ → "show" commands allowed

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config commands denied

Admin B → login → TACACS+ → "show" + configuration commands allowed

Why not RADIUS here?

RADIUS is commonly used for **network access authentication**, such as Wi-Fi, VPN, and 802.1X. TACACS+ is particularly useful when administrators need **fine-grained command authorization on network devices**.

“If a company like Zomato needs to control administrators accessing its Cisco routers and switches, TACACS+ would be suitable because it provides centralized authentication and detailed command-level authorization. It can allow or deny specific commands for different administrators, which is important for controlling and auditing network-device management.”